

ECE 647: Lecture 22

Aloha Random Access

- **Different from the use of Shannon capacity:**
 - Link rate is fixed.
 - Decentralized scheduling.
- Each link attempts transmission with probability p_l .
- Let $N(l)$ be the set of links that interfere with link l .
- The actual probability that link l can send data is:

$$p_l \cdot \prod_{k \in N(l)} (1 - p_k) \quad (1)$$

- Let r_l be the peak-capacity of link l . The optimization problem is:

$$\max \sum_s U_s(X_s) \quad (2)$$

subject to:

$$\sum_s H_s^l X_s \leq r_l p_l \prod_{k \in N(l)} (1 - p_k) \quad (3)$$

- **Transformation of Constraints:**

$$\max \sum_s U_s(X_s) \quad (4)$$

subject to:

$$\log \left(\sum_s H_s^l X_s \right) \leq \log r_l + \log p_l + \sum_{k \in N(l)} \log(1 - p_k) \quad (5)$$

- **Change of Variables:** Let $X_s = e^{y_s}$.

$$\max \sum_s U_s(e^{y_s}) \quad (6)$$

subject to:

$$\log \left(\sum_s H_s^l e^{y_s} \right) \leq \log r_l + \log p_l + \sum_{k \in N(l)} \log(1 - p_k) \quad (7)$$

- The objective can be made concave by suitable choice of the utility functions.
- **Reference:** X. Wang & K. Kar, "Cross-Layer Rate-Control for End-to-End Proportional Fairness in Wireless Networks with Random Access," in ACM Mobihoc, May 2005.

Linear Regression

- **Linear Regression / Signal Estimation**

$$\min_x \|Ax - y\| \quad (8)$$

- **In Signal Estimation:**

- x : Signal/parameter to be estimated.
- y : Measurement, $y = [y_1, y_2, \dots, y_m]^T$.
- A : Design matrix, $A = [a_1, a_2, \dots, a_n]^T$.

- We want:

$$y_i = a_i x \quad (9)$$

- **In Linear Regression:**

- y_i : Dependent variable, or response.
- a_i : Independent variables, or explanatory variables.

Also, we want:

$$y_i = a_i x \quad (10)$$

- The value $y_i - a_i x$ is called the residual (error), e.g., due to noise in the observations.
- **Solution:** (L_2 -norm)

$$\min \|Ax - y\|^2 \triangleq f(x) \quad (11)$$

- **Derivation:**

$$f(x) = (Ax - y)^T (Ax - y) = x^T A^T Ax - 2y^T Ax + y^T y \quad (12)$$

- Gradient of $f(x)$:

$$\nabla f(x) = 2A^T Ax - 2A^T y = 0 \quad (13)$$

- Solving for x :

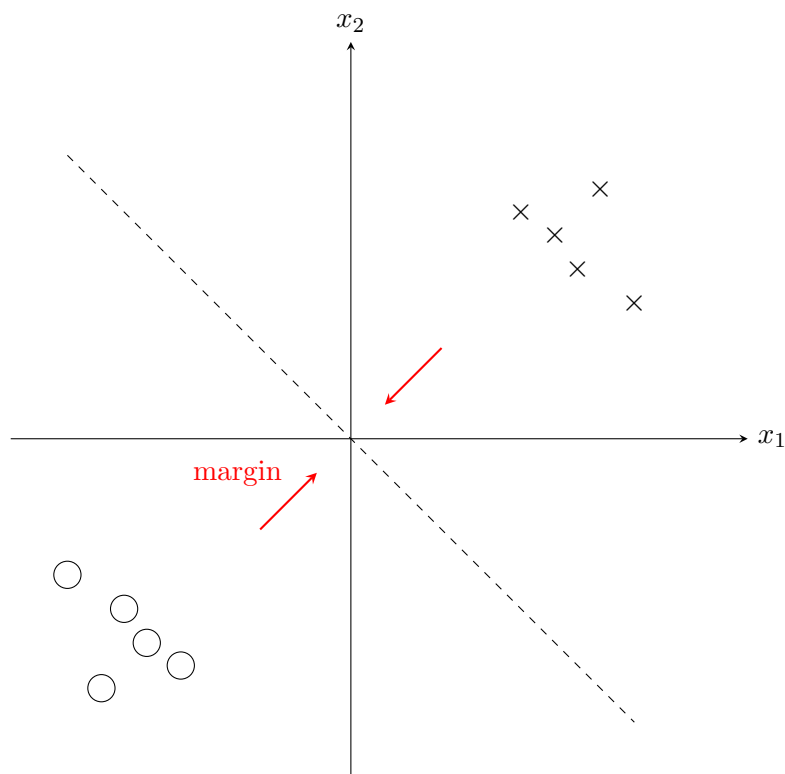
$$\bar{x} = (A^T A)^{-1} A^T y \quad (14)$$

- **References:**

- Boyd, Page 293.
- Other distributions, see Boyd Page 352.

SVM for Classification Problems

- Suppose that we are given a classification problem:
 - (x_i, t_i) , where $i = 1, \dots, n$, represents a dataset of n pairs, where x_i is the independent variable and t_i is the corresponding label.
 - $t_i \in \{-1, 1\}$ (two classes).



- We want to learn a linear model that separates them:

$$y_i = w^T x_i + b \quad (15)$$

- We want:

$$y_i > 0 \iff t_i = 1 \quad (16)$$

$$y_i \leq 0 \iff t_i = -1 \quad (17)$$

- If there exist such (w, b) that satisfy the conditions (16) and (17), the data is said to be linearly separable.
- We may still have many choices for (w, b) :
 - Find the one with the highest margin ("robust").
- For each x_i , the distance to the plane $w^T x + b = 0$ is:

$$\frac{|w^T x_i + b|}{\|w\|_2} \quad (18)$$

- We call this the margin.
- Remove the absolute value sign using t_i :

$$\frac{t_i(w^T x_i + b)}{\|w\|_2} \quad (19)$$

- We therefore want:

$$\max_{w,b} \min_i \frac{t_i(w^T x_i + b)}{\|w\|_2} \quad (20)$$

$$= \max_{w,b} \frac{1}{\|w\|_2} \min_i t_i(w^T x_i + b) \quad (21)$$

- The objective function is not convex in (w, b) , but any (kw, kb) would attain the same value.
- Take (w, b) such that:

$$\min_i t_i(w^T x_i + b) = 1 \quad (22)$$

- We then minimize:

$$\min_{(w,b)} \|w\|_2^2 \quad (23)$$

subject to:

$$\min_i t_i(w^T x_i + b) = 1 \quad (24)$$

- This is equivalent to:

$$\min_{(w,b)} \|w\|_2^2 \quad (25)$$

subject to:

$$\min_i t_i(w^T x_i + b) \geq 1 \quad (26)$$

which is equivalent to:

$$t_i(w^T x_i + b) \geq 1 \quad \text{for all } i. \quad (27)$$

- Those (x_i, t_i) that make:

$$t_i(w^T x_i + b) = 1 \quad (28)$$

are called "support vectors."

- Equivalently, we can:

$$\max_{(w,b)} \min_i t_i(w^T x_i + b) \quad (29)$$

subject to:

$$\|w\|_2^2 \leq 1 \quad (30)$$

- See Boyd P.425 for the non-separable case.

Projection

- Let C be a closed and convex set.
- A point x is said to be the projection of x_0 on C if:

$$\|x - x_0\| = \inf\{\|y - x_0\|, y \in C\} \quad (31)$$

- Later on, when we discuss iterative algorithms for solving convex problems, we will need the projection to the feasible set:

$$\min \|x - x_0\|_2^2 \quad (32)$$

subject to:

$$f_i(x) \leq 0 \quad (33)$$

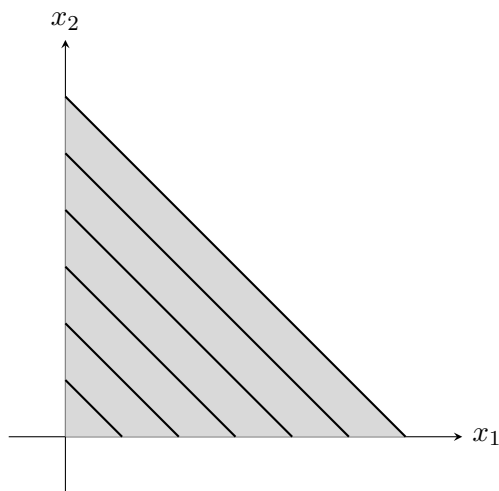
$$h_i(x) = 0 \quad (34)$$

- A convex problem.
- **Projection to a Quadrant:**

$$\min \|x - x_0\|_2^2, \quad \text{subject to } x \geq 0 \quad (35)$$

Solution:

$$x_i = \max(0, x_{0,i}) \quad (36)$$



- **Projection to a Polyhedron (Boyd, Page 390):**

$$\min \|x - x_0\|_2^2, \quad \text{subject to } Ax \leq b \quad (37)$$

This is a quadratic program.

Solution:

- On a hyperplane $a^T x = b$:

$$P_C(x_0) = x_0 + \frac{(b - a^T x_0)a}{\|a\|_2^2} \quad (38)$$

- On a half-space $a^T x \leq b$:

$$P_C(x_0) = \begin{cases} x_0, & \text{if } a^T x_0 \leq b \\ x_0 + \frac{(b - a^T x_0)a}{\|a\|_2^2}, & \text{if } a^T x_0 > b \end{cases} \quad (39)$$

- Can we derive these by optimality conditions?

- **Projection to a Ball:**

$$\min \|x - x_0\|_2^2, \quad \text{subject to } \|x\| \leq r \quad (40)$$

This case is more complicated.